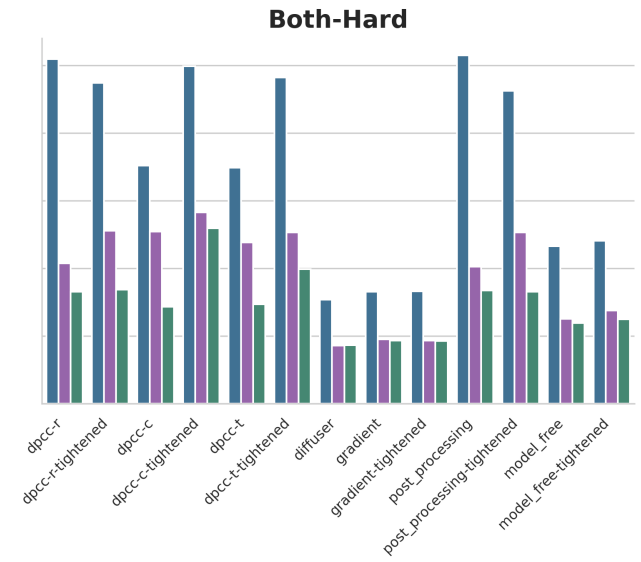
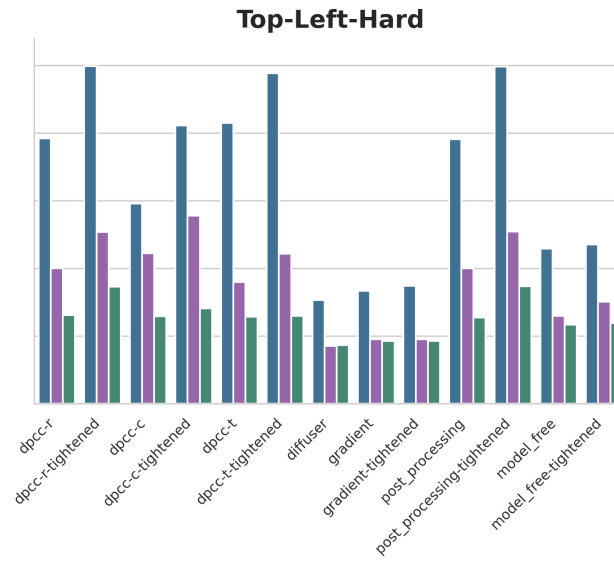
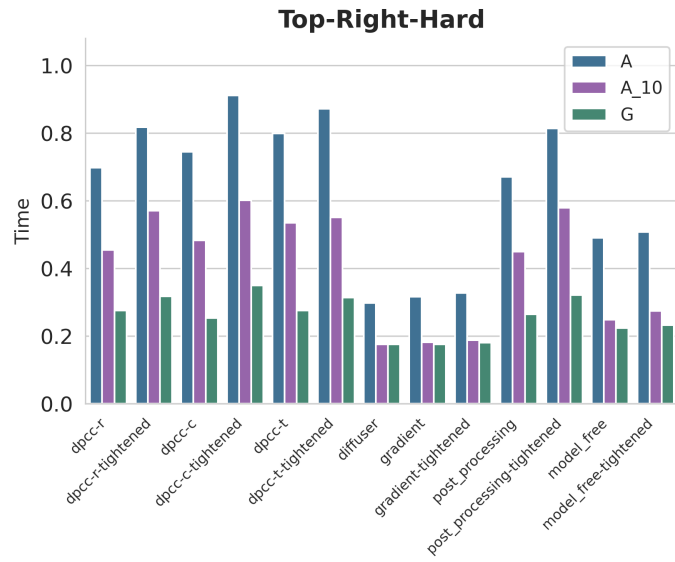


1.2 Computation Time per Step (s)



Section 2 - Complete Raw Data Tables

All metrics for A (Diffusion 20), A_10 (Diffusion 10), and G (FMv3 ODE10).

2.x Top Right Hard

Model A								
Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	66.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.698	-
dpcc-r-tightened	100%	100%	100%	69.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.817	-
dpcc-c	100%	100%	100%	62.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.744	-
dpcc-c-tightened	100%	100%	100%	66.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.911	-
dpcc-t	100%	50%	50%	59.00 +- 3.00	0.50 +- 0.50	0.000 +- 0.000	0.799	-
dpcc-t-tightened	100%	100%	100%	61.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.871	-
diffuser	100%	100%	100%	59.00 +- 3.00	0.00 +- 0.00	0.021 +- 0.002	0.298	0.025
gradient	100%	100%	100%	59.00 +- 3.00	0.00 +- 0.00	0.021 +- 0.002	0.316	-
gradient-tightened	50%	0%	0%	58.00 +- 0.00	15.00 +- 0.00	1.521 +- 0.082	0.327	-
post_processing	100%	100%	100%	66.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.670	-
post_processing-tightened	100%	100%	100%	69.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.814	-
model_free	100%	0%	0%	63.50 +- 2.50	3.00 +- 1.00	0.012 +- 0.011	0.490	-
model_free-tightened	100%	50%	50%	64.00 +- 3.00	2.00 +- 2.00	0.009 +- 0.009	0.508	-
dpcc-c-tightened-dt0p25	100%	100%	100%	81.00 +- 6.00	0.00 +- 0.00	0.002 +- 0.000	2.225	-
dpcc-c-tightened-dt0p5	100%	0%	0%	75.50 +- 2.50	1.00 +- 0.00	0.003 +- 0.000	1.668	-
dpcc-c-tightened-dt2p0	100%	100%	100%	66.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.689	-
dpcc-c-tightened-dt4p0	50%	50%	50%	72.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.535	-

Model A_10								
Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	50%	0%	0%	70.00 +- 0.00	1.50 +- 1.50	0.005 +- 0.005	0.455	-
dpcc-r-tightened	100%	100%	100%	77.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.571	-
dpcc-c	100%	100%	100%	68.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	0.483	-
dpcc-c-tightened	100%	100%	100%	57.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.601	-
dpcc-t	100%	50%	50%	61.00 +- 6.00	1.00 +- 1.00	0.001 +- 0.001	0.535	-

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-t-tightened	100%	100%	100%	56.50 $\pm$ 2.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.551	-
diffuser	100%	0%	0%	63.00 $\pm$ 6.00	17.00 $\pm$ 2.00	1.316 $\pm$ 0.961	0.176	0.031
gradient	50%	0%	0%	69.00 $\pm$ 0.00	18.00 $\pm$ 1.00	1.380 $\pm$ 0.879	0.182	-
gradient-tightened	50%	0%	0%	69.00 $\pm$ 0.00	11.00 $\pm$ 8.00	1.133 $\pm$ 1.077	0.188	-
post_processing	50%	0%	0%	70.00 $\pm$ 0.00	1.50 $\pm$ 1.50	0.005 $\pm$ 0.005	0.450	-
post_processing-tightened	100%	100%	100%	77.50 $\pm$ 0.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.579	-
model_free	50%	0%	0%	70.00 $\pm$ 0.00	10.00 $\pm$ 8.00	1.087 $\pm$ 1.051	0.248	-
model_free-tightened	50%	0%	0%	70.00 $\pm$ 0.00	9.50 $\pm$ 7.50	0.892 $\pm$ 0.854	0.274	-
dpcc-c-tightened-dt0p25	100%	100%	100%	63.50 $\pm$ 4.50	0.00 $\pm$ 0.00	0.001 $\pm$ 0.001	1.166	-
dpcc-c-tightened-dt0p5	100%	100%	100%	69.50 $\pm$ 4.50	0.00 $\pm$ 0.00	0.001 $\pm$ 0.001	0.727	-
dpcc-c-tightened-dt2p0	100%	100%	100%	67.50 $\pm$ 4.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.381	-
dpcc-c-tightened-dt4p0	0%	0%	0%	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.306	-

Model G

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-r	100%	50%	50%	82.00 $\pm$ 19.00	4.50 $\pm$ 4.50	0.045 $\pm$ 0.045	0.275	-
dpcc-r-tightened	100%	100%	100%	87.00 $\pm$ 23.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.318	-
dpcc-c	50%	0%	0%	67.00 $\pm$ 0.00	1.50 $\pm$ 0.50	0.005 $\pm$ 0.004	0.253	-
dpcc-c-tightened	100%	100%	100%	71.00 $\pm$ 5.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.349	-
dpcc-t	50%	0%	0%	65.00 $\pm$ 0.00	2.50 $\pm$ 0.50	0.018 $\pm$ 0.016	0.275	-
dpcc-t-tightened	100%	100%	100%	67.50 $\pm$ 4.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.314	-
diffuser	100%	0%	0%	84.50 $\pm$ 24.50	23.50 $\pm$ 6.50	2.563 $\pm$ 2.077	0.175	0.029
gradient	100%	50%	50%	84.00 $\pm$ 22.00	16.00 $\pm$ 16.00	2.405 $\pm$ 2.386	0.175	-
gradient-tightened	100%	50%	50%	83.50 $\pm$ 21.50	16.00 $\pm$ 16.00	2.337 $\pm$ 2.318	0.180	-
post_processing	100%	50%	50%	82.00 $\pm$ 19.00	4.50 $\pm$ 4.50	0.045 $\pm$ 0.045	0.265	-
post_processing-tightened	100%	100%	100%	87.00 $\pm$ 23.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.321	-
model_free	100%	0%	0%	85.00 $\pm$ 23.00	24.50 $\pm$ 7.50	2.453 $\pm$ 2.092	0.224	-
model_free-tightened	100%	0%	0%	85.50 $\pm$ 23.50	24.50 $\pm$ 7.50	2.334 $\pm$ 1.967	0.232	-
dpcc-c-tightened-dt0p25	100%	100%	100%	78.50 $\pm$ 3.50	0.00 $\pm$ 0.00	0.002 $\pm$ 0.000	0.681	-
dpcc-c-tightened-dt0p5	100%	100%	100%	77.50 $\pm$ 5.50	0.00 $\pm$ 0.00	0.001 $\pm$ 0.000	0.426	-
dpcc-c-tightened-dt2p0	100%	100%	100%	71.00 $\pm$ 6.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.272	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-c-tightened-dt4p0	100%	100%	100%	131.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.270	-

2.x Top Left Hard

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	64.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.784	-
dpcc-r-tightened	100%	100%	100%	65.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.998	-
dpcc-c	100%	50%	50%	62.50 +- 4.50	1.00 +- 1.00	0.003 +- 0.003	0.592	-
dpcc-c-tightened	100%	100%	100%	63.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.822	-
dpcc-t	100%	50%	50%	61.50 +- 3.50	1.50 +- 1.50	0.004 +- 0.004	0.829	-
dpcc-t-tightened	100%	100%	100%	60.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.976	-
diffuser	100%	0%	0%	59.00 +- 3.00	20.00 +- 2.00	2.515 +- 0.387	0.306	0.025
gradient	100%	0%	0%	58.50 +- 3.50	17.50 +- 3.50	2.116 +- 0.571	0.334	-
gradient-tightened	100%	0%	0%	59.00 +- 3.00	19.50 +- 1.50	2.218 +- 0.316	0.348	-
post_processing	100%	100%	100%	64.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.782	-
post_processing-tightened	100%	100%	100%	65.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.996	-
model_free	100%	0%	0%	63.50 +- 2.50	19.00 +- 0.00	2.829 +- 0.455	0.458	-
model_free-tightened	100%	0%	0%	64.50 +- 3.50	15.00 +- 2.00	2.029 +- 0.233	0.470	-
dpcc-c-tightened-dt0p25	100%	100%	100%	70.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	1.259	-
dpcc-c-tightened-dt0p5	100%	100%	100%	63.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.937	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.679	-
dpcc-c-tightened-dt4p0	100%	0%	0%	107.00 +- 2.00	16.50 +- 0.50	3.711 +- 0.053	0.810	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	68.50 +- 4.50	2.50 +- 2.50	0.047 +- 0.047	0.400	-
dpcc-r-tightened	100%	100%	100%	66.50 +- 5.50	0.00 +- 0.00	0.000 +- 0.000	0.507	-
dpcc-c	100%	0%	0%	86.00 +- 8.00	6.50 +- 4.50	0.102 +- 0.034	0.444	-
dpcc-c-tightened	100%	100%	100%	58.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.556	-
dpcc-t	100%	0%	0%	95.50 +- 2.50	24.00 +- 2.00	0.417 +- 0.034	0.360	-
dpcc-t-tightened	100%	100%	100%	55.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.443	-

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
diffuser	100%	0%	0%	63.00 $\pm$ 6.00	27.00 $\pm$ 2.00	5.820 $\pm$ 4.193	0.171	0.031
gradient	100%	0%	0%	67.50 $\pm$ 1.50	30.00 $\pm$ 5.00	6.548 $\pm$ 5.100	0.190	-
gradient-tightened	100%	0%	0%	67.00 $\pm$ 2.00	29.50 $\pm$ 4.50	6.110 $\pm$ 4.702	0.190	-
post_processing	100%	50%	50%	68.50 $\pm$ 4.50	2.50 $\pm$ 2.50	0.047 $\pm$ 0.047	0.400	-
post_processing-tightened	100%	100%	100%	66.50 $\pm$ 5.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.509	-
model_free	100%	0%	0%	107.50 $\pm$ 35.50	66.50 $\pm$ 40.50	17.060 $\pm$ 15.519	0.260	-
model_free-tightened	100%	0%	0%	76.00 $\pm$ 3.00	35.00 $\pm$ 8.00	6.907 $\pm$ 5.334	0.301	-
dpcc-c-tightened-dt0p25	100%	50%	50%	72.50 $\pm$ 4.50	1.50 $\pm$ 1.50	0.008 $\pm$ 0.007	0.997	-
dpcc-c-tightened-dt0p5	100%	100%	100%	65.50 $\pm$ 2.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.731	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.50 $\pm$ 5.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.403	-
dpcc-c-tightened-dt4p0	100%	0%	0%	123.50 $\pm$ 2.50	11.00 $\pm$ 6.00	2.219 $\pm$ 1.589	0.421	-

Model G

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-r	100%	100%	100%	75.50 $\pm$ 13.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.262	-
dpcc-r-tightened	100%	100%	100%	77.00 $\pm$ 13.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.346	-
dpcc-c	100%	100%	100%	69.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.258	-
dpcc-c-tightened	100%	100%	100%	67.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.282	-
dpcc-t	100%	50%	50%	94.00 $\pm$ 32.00	22.00 $\pm$ 22.00	0.250 $\pm$ 0.250	0.257	-
dpcc-t-tightened	100%	100%	100%	64.50 $\pm$ 2.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.259	-
diffuser	100%	0%	0%	84.50 $\pm$ 24.50	24.50 $\pm$ 10.50	2.978 $\pm$ 2.559	0.173	0.029
gradient	100%	0%	0%	85.00 $\pm$ 24.00	16.00 $\pm$ 2.00	1.391 $\pm$ 0.993	0.185	-
gradient-tightened	100%	0%	0%	85.00 $\pm$ 24.00	16.00 $\pm$ 2.00	1.351 $\pm$ 0.943	0.185	-
post_processing	100%	100%	100%	75.50 $\pm$ 13.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.254	-
post_processing-tightened	100%	100%	100%	77.00 $\pm$ 13.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.347	-
model_free	100%	0%	0%	86.00 $\pm$ 24.00	25.50 $\pm$ 12.50	3.278 $\pm$ 2.909	0.234	-
model_free-tightened	100%	0%	0%	86.00 $\pm$ 24.00	25.50 $\pm$ 11.50	2.773 $\pm$ 2.399	0.239	-
dpcc-c-tightened-dt0p25	100%	100%	100%	76.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.529	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 $\pm$ 2.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.329	-
dpcc-c-tightened-dt2p0	100%	100%	100%	67.00 $\pm$ 3.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.268	-
dpcc-c-tightened-dt4p0	100%	100%	100%	124.50 $\pm$ 4.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.254	-

2.x Both Hard

Model A

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-r	100%	0%	0%	61.50 $\pm$ 0.50	2.50 $\pm$ 0.50	0.002 $\pm$ 0.001	1.019	-
dpcc-r-tightened	100%	100%	100%	66.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.948	-
dpcc-c	100%	0%	0%	84.50 $\pm$ 11.50	19.00 $\pm$ 17.00	0.381 $\pm$ 0.377	0.704	-
dpcc-c-tightened	100%	100%	100%	56.50 $\pm$ 2.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.997	-
dpcc-t	100%	50%	50%	81.50 $\pm$ 18.50	17.00 $\pm$ 17.00	0.416 $\pm$ 0.416	0.697	-
dpcc-t-tightened	100%	100%	100%	62.50 $\pm$ 2.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.964	-
diffuser	100%	0%	0%	59.00 $\pm$ 3.00	7.50 $\pm$ 0.50	0.090 $\pm$ 0.001	0.307	0.025
gradient	100%	0%	0%	59.00 $\pm$ 3.00	7.00 $\pm$ 1.00	0.064 $\pm$ 0.007	0.331	-
gradient-tightened	100%	0%	0%	61.00 $\pm$ 3.00	6.00 $\pm$ 1.00	0.044 $\pm$ 0.004	0.332	-
post_processing	100%	0%	0%	61.50 $\pm$ 0.50	2.50 $\pm$ 0.50	0.002 $\pm$ 0.001	1.029	-
post_processing-tightened	100%	100%	100%	66.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.924	-
model_free	100%	0%	0%	63.00 $\pm$ 2.00	8.50 $\pm$ 0.50	0.103 $\pm$ 0.008	0.465	-
model_free-tightened	100%	0%	0%	63.50 $\pm$ 2.50	8.50 $\pm$ 0.50	0.091 $\pm$ 0.010	0.481	-
dpcc-c-tightened-dt0p25	100%	50%	50%	75.50 $\pm$ 0.50	0.50 $\pm$ 0.50	0.000 $\pm$ 0.000	1.835	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 $\pm$ 4.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	1.185	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.50 $\pm$ 2.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.737	-
dpcc-c-tightened-dt4p0	100%	100%	100%	85.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.691	-

Model A_10

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-r	100%	0%	0%	74.00 $\pm$ 3.00	2.00 $\pm$ 1.00	0.005 $\pm$ 0.004	0.415	-
dpcc-r-tightened	100%	100%	100%	71.50 $\pm$ 7.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.511	-
dpcc-c	100%	50%	50%	79.50 $\pm$ 18.50	23.00 $\pm$ 23.00	0.490 $\pm$ 0.490	0.509	-
dpcc-c-tightened	100%	100%	100%	63.50 $\pm$ 1.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.565	-
dpcc-t	100%	50%	50%	70.00 $\pm$ 6.00	1.00 $\pm$ 1.00	0.043 $\pm$ 0.043	0.477	-
dpcc-t-tightened	100%	100%	100%	69.50 $\pm$ 1.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.506	-
diffuser	100%	0%	0%	63.00 $\pm$ 6.00	8.50 $\pm$ 3.50	1.415 $\pm$ 1.363	0.172	0.031
gradient	100%	0%	0%	63.50 $\pm$ 5.50	10.00 $\pm$ 5.00	1.335 $\pm$ 1.297	0.190	-
gradient-tightened	100%	0%	0%	62.00 $\pm$ 7.00	9.00 $\pm$ 5.00	1.463 $\pm$ 1.432	0.187	-

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
post_processing	100%	0%	0%	74.00 $\pm$ 3.00	2.00 $\pm$ 1.00	0.005 $\pm$ 0.004	0.405	-
post_processing-tightened	100%	100%	100%	71.50 $\pm$ 7.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.506	-
model_free	100%	0%	0%	76.50 $\pm$ 6.50	4.50 $\pm$ 1.50	0.038 $\pm$ 0.001	0.251	-
model_free-tightened	100%	0%	0%	64.50 $\pm$ 5.50	9.50 $\pm$ 3.50	1.394 $\pm$ 1.360	0.276	-
dpcc-c-tightened-dt0p25	100%	50%	50%	74.50 $\pm$ 3.50	1.00 $\pm$ 1.00	0.002 $\pm$ 0.002	0.875	-
dpcc-c-tightened-dt0p5	100%	100%	100%	64.50 $\pm$ 3.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.676	-
dpcc-c-tightened-dt2p0	100%	100%	100%	62.50 $\pm$ 3.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.398	-
dpcc-c-tightened-dt4p0	100%	100%	100%	94.00 $\pm$ 3.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.379	-

Model G

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-r	100%	50%	50%	92.00 $\pm$ 32.00	35.50 $\pm$ 35.50	0.478 $\pm$ 0.478	0.331	-
dpcc-r-tightened	100%	100%	100%	57.50 $\pm$ 0.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.337	-
dpcc-c	100%	0%	0%	138.00 $\pm$ 12.00	55.50 $\pm$ 8.50	0.598 $\pm$ 0.276	0.286	-
dpcc-c-tightened	100%	100%	100%	58.50 $\pm$ 3.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.519	-
dpcc-t	100%	0%	0%	135.00 $\pm$ 4.00	65.50 $\pm$ 0.50	0.736 $\pm$ 0.074	0.294	-
dpcc-t-tightened	100%	100%	100%	64.00 $\pm$ 1.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.398	-
diffuser	100%	50%	50%	84.50 $\pm$ 24.50	2.00 $\pm$ 2.00	0.026 $\pm$ 0.010	0.173	0.029
gradient	100%	50%	50%	84.50 $\pm$ 23.50	2.00 $\pm$ 2.00	0.023 $\pm$ 0.005	0.187	-
gradient-tightened	100%	0%	0%	81.00 $\pm$ 20.00	8.50 $\pm$ 5.50	1.602 $\pm$ 1.581	0.186	-
post_processing	100%	50%	50%	92.00 $\pm$ 32.00	35.50 $\pm$ 35.50	0.478 $\pm$ 0.478	0.335	-
post_processing-tightened	100%	100%	100%	57.50 $\pm$ 0.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.331	-
model_free	100%	50%	50%	85.00 $\pm$ 23.00	2.00 $\pm$ 2.00	0.006 $\pm$ 0.006	0.239	-
model_free-tightened	100%	50%	50%	85.50 $\pm$ 23.50	1.50 $\pm$ 1.50	0.003 $\pm$ 0.003	0.250	-
dpcc-c-tightened-dt0p25	100%	100%	100%	70.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.666	-
dpcc-c-tightened-dt0p5	100%	100%	100%	62.00 $\pm$ 5.00	0.00 $\pm$ 0.00	0.001 $\pm$ 0.001	0.449	-
dpcc-c-tightened-dt2p0	100%	100%	100%	72.50 $\pm$ 8.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.323	-
dpcc-c-tightened-dt4p0	100%	100%	100%	101.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.297	-

Section 3 - Three-Way Head-to-Head Comparison

Best G&C highlighted green. Lowest violation count highlighted teal. $\Delta(A_{10}-A)$ tracks the direct effect of halving diffusion steps.

3.x Top Right Hard

Method	A G&C	A_10 G&C	G G&C	Best	$\Delta(A_{10}-A)$	A Viol	A_10 Viol	G Viol	A Time	A_10 Time	G Time
dpcc-r	100%	0%	50%	A	-100pp	0.0	1.5	4.5	0.698	0.455	0.275
dpcc-r-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.817	0.571	0.318
dpcc-c	100%	100%	0%	A/A_10	=	0.0	0.0	1.5	0.744	0.483	0.253
dpcc-c-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.911	0.601	0.349
dpcc-t	50%	50%	0%	A/A_10	=	0.5	1.0	2.5	0.799	0.535	0.275
dpcc-t-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.871	0.551	0.314
diffuser	100%	0%	0%	A	-100pp	0.0	17.0	23.5	0.298	0.176	0.175
gradient	100%	0%	50%	A	-100pp	0.0	18.0	16.0	0.316	0.182	0.175
gradient-tightened	0%	0%	50%	G	=	15.0	11.0	16.0	0.327	0.188	0.180
post_processing	100%	0%	50%	A	-100pp	0.0	1.5	4.5	0.670	0.450	0.265
post_processing-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.814	0.579	0.321
model_free	0%	0%	0%	A/A_10/G	=	3.0	10.0	24.5	0.490	0.248	0.224
model_free-tightened	50%	0%	0%	A	-50pp	2.0	9.5	24.5	0.508	0.274	0.232
dpcc-c-tightened-dt0p25	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	2.225	1.166	0.681
dpcc-c-tightened-dt0p5	0%	100%	100%	A_10/G	+100pp	1.0	0.0	0.0	1.668	0.727	0.426
dpcc-c-tightened-dt2p0	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.689	0.381	0.272
dpcc-c-tightened-dt4p0	50%	0%	100%	G	-50pp	0.0	0.0	0.0	0.535	0.306	0.270

3.x Top Left Hard

Method	A G&C	A_10 G&C	G G&C	Best	$\Delta(A_{10}-A)$	A Viol	A_10 Viol	G Viol	A Time	A_10 Time	G Time
dpcc-r	100%	50%	100%	A/G	-50pp	0.0	2.5	0.0	0.784	0.400	0.262
dpcc-r-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.998	0.507	0.346
dpcc-c	50%	0%	100%	G	-50pp	1.0	6.5	0.0	0.592	0.444	0.258
dpcc-c-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.822	0.556	0.282
dpcc-t	50%	0%	50%	A/G	-50pp	1.5	24.0	22.0	0.829	0.360	0.257
dpcc-t-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.976	0.443	0.259

Method	A G&C	A_10 G&C	G G&C	Best	$\Delta(A_{10}-A)$	A Viol	A_10 Viol	G Viol	A Time	A_10 Time	G Time
diffuser	0%	0%	0%	A/A_10/G	=	20.0	27.0	24.5	0.306	0.171	0.173
gradient	0%	0%	0%	A/A_10/G	=	17.5	30.0	16.0	0.334	0.190	0.185
gradient-tightened	0%	0%	0%	A/A_10/G	=	19.5	29.5	16.0	0.348	0.190	0.185
post_processing	100%	50%	100%	A/G	-50pp	0.0	2.5	0.0	0.782	0.400	0.254
post_processing-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.996	0.509	0.347
model_free	0%	0%	0%	A/A_10/G	=	19.0	66.5	25.5	0.458	0.260	0.234
model_free-tightened	0%	0%	0%	A/A_10/G	=	15.0	35.0	25.5	0.470	0.301	0.239
dpcc-c-tightened-dt0p25	100%	50%	100%	A/G	-50pp	0.0	1.5	0.0	1.259	0.997	0.529
dpcc-c-tightened-dt0p5	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.937	0.731	0.329
dpcc-c-tightened-dt2p0	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.679	0.403	0.268
dpcc-c-tightened-dt4p0	0%	0%	100%	G	=	16.5	11.0	0.0	0.810	0.421	0.254

3.x Both Hard

Method	A G&C	A_10 G&C	G G&C	Best	$\Delta(A_{10}-A)$	A Viol	A_10 Viol	G Viol	A Time	A_10 Time	G Time
dpcc-r	0%	0%	50%	G	=	2.5	2.0	35.5	1.019	0.415	0.331
dpcc-r-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.948	0.511	0.337
dpcc-c	0%	50%	0%	A_10	+50pp	19.0	23.0	55.5	0.704	0.509	0.286
dpcc-c-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.997	0.565	0.519
dpcc-t	50%	50%	0%	A/A_10	=	17.0	1.0	65.5	0.697	0.477	0.294
dpcc-t-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.964	0.506	0.398
diffuser	0%	0%	50%	G	=	7.5	8.5	2.0	0.307	0.172	0.173
gradient	0%	0%	50%	G	=	7.0	10.0	2.0	0.331	0.190	0.187
gradient-tightened	0%	0%	0%	A/A_10/G	=	6.0	9.0	8.5	0.332	0.187	0.186
post_processing	0%	0%	50%	G	=	2.5	2.0	35.5	1.029	0.405	0.335
post_processing-tightened	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.924	0.506	0.331
model_free	0%	0%	50%	G	=	8.5	4.5	2.0	0.465	0.251	0.239
model_free-tightened	0%	0%	50%	G	=	8.5	9.5	1.5	0.481	0.276	0.250
dpcc-c-tightened-dt0p25	50%	50%	100%	G	=	0.5	1.0	0.0	1.835	0.875	0.666
dpcc-c-tightened-dt0p5	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	1.185	0.676	0.449
dpcc-c-tightened-dt2p0	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.737	0.398	0.323

Method	A G&C	A_10 G&C	G G&C	Best	$\Delta(A_{10}-A)$	A Viol	A_10 Viol	G Viol	A Time	A_10 Time	G Time
dpcc-c-tightened-dt4p0	100%	100%	100%	A/A_10/G	=	0.0	0.0	0.0	0.691	0.379	0.297

Section 5 - Mathematical Summary & Findings

1. The Discretization Error Trade-off in Diffusion

Diffusion models generate samples by integrating a reverse stochastic differential equation (SDE). By halving the integration steps from 20 (A) to 10 (A_10), we double the temporal step size Δt . Mathematically, this increases the local truncation error of the discrete solver at each step. Because the score function field ($\nabla_x \log p_t(x)$) tends to be highly curved, large Δt steps cause the trajectory to drift away from the true manifold.

This is empirically visible in the data: A_10 suffers severe success rate drops in baseline methods. For instance, in top-right-hard, the diffuser success drops from 100% (A) to 0% (A_10), accumulating 17 violations per path. Diffusion models fundamentally require small step sizes to navigate complex geometric constraints safely.

2. Vector Field Linearity: Flow Matching vs. Diffusion

Flow Matching (G), configured with ODE=10, intrinsically formulates probability paths using continuous, straight-line target vector fields. Because the learned vector field $v_\theta(x, t)$ is inherently less curved than the diffusion score field, it incurs significantly lower integration error for a given Δt . Consequently, G easily outperforms A_10 across most baselines and untightened scenarios despite operating on the exact same 10-step compute budget.

3. Computational Symmetry & Projection Efficacy

As expected mathematically, halving the diffusion steps scales the raw compute time linearly. Baseline methods for A_10 clock in at ~0.17-0.19s, perfectly matching G's baseline times, effectively normalizing the compute budget between the two architectures.

Interestingly, the DPCC tightened methods completely mask the integration error for both A_10 and G. Once the tightened projection forces constraint satisfaction at every internal step, A_10 restores its 100% G&C success rate across all hard scenarios, albeit at a higher per-step time cost than the baselines.

4. Final Verdict

If constrained to a 10-step computational budget, **G (FMv3)** is geometrically and mathematically superior to **A_10 (Diffusion)**. The straight paths of Flow Matching survive the discretization penalty of a large Δt far better than the curved paths of diffusion. A_10 can only remain viable if combined with robust (and expensive) tightened projection methods.